Dylan Khader

Engineering student | Electrical, electronic and mechanical design

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Profile

I'm an engineering student at Ron Dearing UTC with a strong interest in electrical and electronic engineering, particularly how circuits and mechanisms work together to make things move. My Level 2 electrical installation qualification and projects involving CAD, 3D printing and assembly have helped me connect theory with practical work. I enjoy designing, building and improving my own projects, with a longer-term ambition to develop robotic systems.

Engineering projects

RC car design and build

Independent project | CAD, 3D printing and assembly

- Designed my own chassis and steering system in Autodesk Inventor, then 3D printed the parts and assembled the car with motors and radio-control electronics.
- Built a second version with a larger steering gear and guides to keep the steering bar aligned. Adjusted the motor mount and pulley supports to improve fit.
- Scaled down and modified a downloaded body model to fit my chassis.

Wind turbine CAD, drawings and animation

Ron Dearing UTC | Unit 10 CAD coursework

- Modelled a multi-part wind turbine in Autodesk Inventor, using lofted blade profiles and rigid and rotational assembly joints.
- Produced dimensioned component and assembly drawings, an exploded assembly animation and an animated scene showing rotor rotation.

Greenpower steering wheel and accelerator

Ron Dearing UTC | CAD design project

- Developed concepts using class hand measurements, comparing materials and manufacturing options.
- Created Inventor assemblies, exploded views and drawings, considering control positioning, access for fasteners and ease of assembly.

Home electrical containment proposal

Independent design project | CAD and materials planning

- After noticing loose external wiring at home, I used my electrical training and CAD skills to develop a containment proposal. Modelled the house, planned routes and prepared a materials list and presentation showing the proposed layout.

Education

Ron Dearing UTC, Hull

Level 3 Engineering programme | OCR double course and EAL diploma

OCR Engineering: expected completion April 2027; predicted Distinction*. Achieved Distinctions in Mathematics for Engineering, Science for Engineering, Mechanical Design, Computer Aided Design, Mechanical Operations and Computer Aided Manufacturing (2026).

EAL Level 3 Subsidiary Diploma in Engineering: Merit (2026).

Hull College

City & Guilds Level 2 Technical Certificate in Electrical Installation | 2025

Merit overall. Distinction in the theory exam.

Experience and practical skills

Work experience

Electrical work experience

The Gas Company UK Ltd, Hull | One week, January 2025

Selected by my Hull College tutor for a placement shadowing an electrician on domestic visits. Gained supervised experience of fault finding, electrical accessories and condition reporting. Written feedback praised my punctuality, attitude and conscientious work in customers' homes.

Crew Member

McDonald's, Hull | May 2025 to present

Work as part of a busy restaurant team alongside full-time study. Used to staying focused under pressure, communicating with colleagues and contributing during demanding shifts.

Front of House Assistant

Castello Pizzeria, Beverley | October 2025 to present

Take customer orders, pack meals and prepare orders for delivery in a part-time role helping my uncle alongside McDonalds and college.

Technical skills

- CAD and drawings: Autodesk Inventor part modelling, lofts, assembly joints, exploded views, animations and dimensioned drawings. AutoCAD 2D drawing and revision exercises.
- 3D printing: CAD export to STL, preparation in UltiMaker Cura and practical printed components for coursework and my own RC car.
- CNC and CAM: manual G-code work, toolpath simulation and workshop setup using VR Milling. Calculated hole coordinates and checked finished dimensions.
- Workshop practice: supervised use of centre lathes, milling machines and pillar drills; marking out, drilling, hand tapping, turning and assembly.
- Measurement and quality: reading drawings, using vernier callipers and height gauges, checking tolerances and recording deviations to identify where a part needs improvement.

Personal projects and interests

Building a heart-shaped LED chaser from individual components, using NE555 timers and CD4017 decade counters. The project is in progress and is helping me apply electrical theory through practical circuit assembly.

Most of my spare time goes into creating projects. I also enjoy going to the gym, spending time with friends and reading.

Recognition

Hull College Most Improved Person of the Year, 2025.

Kingswood Academy Outstanding Attainment in Physical Education, 2023.